

# Faezeh Manesh

Department of Physics and Astronomy, University of California, Riverside

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## Research Interests

**Observational Galaxy Formation and Evolution:** massive galaxies, quenching of star formation, stellar mass assembly, spatially-resolved stellar populations, first galaxies

## Education

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|-------------|----------------------------------------------------------------|----------------------|
| 2023 – 2028 | <b>Ph.D. in Astronomy, University of California, Riverside</b> | <i>Riverside, CA</i> |
| 2025        | <b>M.Sc. in Astronomy, University of California, Riverside</b> | <i>Riverside, CA</i> |
| 2022        | <b>B.Sc. in Physics, Shiraz University</b>                     | <i>Shiraz, Iran</i>  |

## Positions

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|----------------|-----------------------------------------------------------------------------------------------|----------------------|
| 2023 – Present | <b>Graduate Research Assistant, University of California, Riverside</b>                       | <i>Riverside, CA</i> |
| 2022 – 2023    | <b>Undergraduate Research Assistant, Institute for Research in Fundamental Sciences (IPM)</b> | <i>Tehran, Iran</i>  |
| 2018 – 2022    | <b>Undergraduate Research Assistant, Biruni Observatory, Shiraz University</b>                | <i>Shiraz, Iran</i>  |

## Publications

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| 2026 | <b>Unveiling Fast-Forming Galaxies at Cosmic Dawn (In preparation)</b><br>F. Manesh, J. Weaver, B. Siana, ..., et al.                                                                      |
| 2026 | <b>Spatially Resolved Nebular-Stellar Reddening with JWST/NIRISS</b><br>P. J. Watson, ..., including F. Manesh, et al.                                                                     |
| 2026 | <b>The Fraction of Clumpy Galaxies in JWST Surveys over <math>2 &lt; z &lt; 12</math></b><br>A. De la Vega, ..., including F. Manesh, et al.                                               |
| 2023 | <b>COSMOS2020: Disentangling the Role of Mass and Environment in Star Formation Activity of Galaxies at <math>0.4 &lt; z &lt; 4</math></b><br>S. Taamoli, ..., including F. Manesh, et al. |

## Research Projects

### PhD Research

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| 2026 – Present | <b>Characterizing Bursty Star Formation Histories of Dwarf Galaxies at <math>z = 1-2.4</math> in the Abell 2744 Lensing Field</b><br>JWST Cycle 3 AR Proposal 6319 · PI: Dr. Anahita Alavi |
| 2023 – Present | <b>Star Formation Histories and Spatially Resolved Properties of Ultraluminous <math>z &gt; 7</math> Galaxies</b><br>JWST Cycle 1 GO2659 project · PI: Dr. John R. Weaver                  |

### Undergraduate Research

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| 2022 – 2023 | <b>Star Formation History in a Dwarf Galaxy, NGC 6822</b><br>Supervisor: Dr. Atefeh Javadi · Institute for Research in Fundamental Sciences (IPM) |
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| 2021 – 2022 | <b>Observation of Transiting Exoplanets: Transit Light Curves of the Hot Jupiter HAT-P-32b</b><br>Supervisor: Dr. Moein Mosleh · Shiraz University |
| Spring 2020 | <b>Asteroid Detection with Small Telescopes</b>                                                                                                    |

## Talks & Posters

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| Aug. 2026 | <b>TALK</b><br><b>Constraining Star-Formation Burstiness in Lensed Dwarf Galaxies at <math>z = 0.5\text{--}2.3</math> with HST and GLASS-JWST/NIRISS</b><br>Greater IPAC Science Symposium                                    | <i>Pasadena, CA</i>        |
| Aug. 2026 | <b>POSTER</b><br><b>Constraining Star-Formation Burstiness in Lensed Dwarf Galaxies at <math>z = 0.5\text{--}2.3</math> with HST and GLASS-JWST/NIRISS</b><br>Surveying the Universe in 4D, Space Telescope Science Institute | <i>Baltimore, MD</i>       |
| Jun. 2026 | <b>POSTER</b><br><b>Massive galaxy growth in action: A major merger at <math>z = 7.7</math></b><br>248th of the American Astronomical Society                                                                                 | <i>Pasadena, CA</i>        |
| Mar. 2026 | <b>POSTER</b><br><b>Massive galaxy growth in action: A major merger at <math>z = 7.7</math></b><br>Aspen meeting                                                                                                              | <i>Aspen, CO</i>           |
| Oct. 2025 | <b>CONFERENCE TALK</b><br><b>Unveiling Fast-Forming Galaxies at Cosmic Dawn</b><br>Infrared Spectroscopy from Space, IPAC                                                                                                     | <i>Pasadena, CA</i>        |
| Aug. 2025 | <b>POSTER</b><br><b>Unveiling Fast-Forming Galaxies at Cosmic Dawn</b><br>Santa Cruz Galaxy Workshop                                                                                                                          | <i>Santa Cruz, CA</i>      |
| Jan. 2025 | <b>CONFERENCE TALK</b><br><b>Unveiling Fast-Forming Galaxies at Cosmic Dawn</b><br>245th Meeting of the American Astronomical Society                                                                                         | <i>National Harbor, MD</i> |
| Jan. 2023 | <b>SEMINAR</b><br><b>Observation of Transiting Exoplanet with Small Telescopes</b><br>Institute for Research in Fundamental Sciences                                                                                          | <i>IPM, Tehran</i>         |

## Awards & Honors

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| 2026 | <b>Martin and Beate Block Award</b><br>Aspen Center for Physics               |
| 2023 | <b>Dean's Distinguished Fellowship</b><br>University of California, Riverside |
| 2022 | <b>Ranked 6<sup>th</sup> among 30 Physics Students</b><br>Shiraz University   |
| 2017 | <b>Tuition-Waiver Scholarship</b><br>Shiraz University                        |

### Travel Awards

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| 2026 | <b>UCR Graduate Student Association \$600 Conference Travel Grant</b><br>University of California, Riverside |
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## Observing Experience

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| Spring 2025 | <b>Lick Observatory Workshop</b><br>Observational training using the Shane (3m) and Nickel (1m) telescopes through the UCO Observational Astronomy Workshop |
| Fall 2021   | <b>Transit Photometry of HAT-P-32b</b><br>Biruni Observatory 20-inch telescope · 2 nights                                                                   |

## Teaching Experience

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| 2023 – 2028 | <b>Teaching Assistant, University of California, Riverside</b><br>Laboratory of General Physics — Spring 2024, Fall 2024, Winter 2025                                                                                                                |
| 2018 – 2022 | <b>Teaching Assistant, Shiraz University</b><br>General Physics 111 — Spring 2021<br>Introduction to Astronomy and Astrophysics — Fall 2018, Fall 2021, Fall 2022<br>Prepared homework problems, graded assignments, and taught discussion sections. |

## Skills

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### Data Analysis

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| <b>Photometry</b>          | Aperture & PSF photometry (SourceExtractor, Photutils), SED fitting (EAZY, Bagpipes, Prospector)         |
| <b>Spectroscopy</b>        | Integral field spectroscopy, grism spectroscopy; emission-line measurements, stellar population modeling |
| <b>Morphology</b>          | Surface-brightness profile fitting (GALFIT, PySersic)                                                    |
| <b>Computing Languages</b> | Python (NumPy, SciPy, Astropy, Matplotlib), HTML, CSS                                                    |
| <b>Tools</b>               | Git, Linux/Unix, L <sup>A</sup> T <sub>E</sub> X                                                         |

## Service & Public Outreach

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| Summer 2026    | <b>Journal Club Coordinator, UC Riverside</b><br>Organized weekly astronomy journal club meetings and paper discussions.                                                                                                                    |
| 2024 – Present | <b>Core Member, AURORA: Astronomy Union for Reading, Outreach, and Research Activities</b><br>Actively involved in organizing events and activities to foster a vibrant and engaging environment for graduate students in astronomy at UCR. |
| 2024 – Present | <b>Co-manager &amp; Author, Staryab Website</b><br>Write monthly literature summaries, edit contributions, and coordinate communication among Iranian professional astronomers worldwide for a Farsi-language astrophysics platform.        |
| 2023 – Present | <b>Stargazing Outreach Volunteer, University of California, Riverside</b><br>Guide public telescope viewing and share astronomy concepts with students, children, and the local community.                                                  |
| 2020 – 2021    | <b>Astronomy Society President, Shiraz University</b>                                                                                                                                                                                       |
| 2018 – 2022    | <b>Stargazing Outreach Volunteer &amp; Core Member, Biruni Observatory, Shiraz University</b><br>Core member responsible for managing public events and outreach programs at the observatory.                                               |
| 2018 – 2020    | <b>Public Lecturer, Biruni Observatory, Shiraz University, Iran</b><br>Gave introductory talks on astronomy and night-sky observing to general audiences at the observatory.                                                                |

## Diversity, Equity, & Inclusion

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- 2026      **Participant, Chancellor's Making Excellence Inclusive (MEI) Diversity Certificate Program, University of California, Riverside**  
Completed a multi-week certificate program on diversity, equity, and inclusion through workshops and discussion sessions; authored a white paper and designed and presented a capstone project applying DEI concepts to the university environment.
- 2025 – 2026      **President, Physics Organization for Women and Underrepresented at UCR (POWUR)**  
Lead an organization dedicated to supporting and advocating for women and underrepresented students in the Department of Physics and Astronomy; organize community-building, mentoring, and professional-development events.

## Languages

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Farsi (Native)      English (Fluent)

## Hobbies

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Night sky photography, pottery, knitting, and random arts & crafts